Tracing Cross-Field Responses to Input Perturbations in Transformer-Based Weather Forecasting Models

Anonymous Author(s)
Affiliation
Address
email

Abstract

Aggregate forecast scores do not show how an error in one atmospheric field changes the rest of a multivariate forecast. We audit this behavior in two transformer-based weather models, Aurora and Pangu-Weather, by replacing each of 19 inputs with season- and UTC-matched climatology while holding the other fields fixed. Paired rollouts over 48 dates produce directed 19×19 response matrices at +6 and +24 h, supplemented by six-dose sweeps and grid-cell error maps. In both models, replacing $Z850$ changes $V850$ about nine times more than the reverse intervention. The diagonal $Z500$ displacement falls by roughly half from +6 to +24 h, while responses in other fields persist. Lower-level geopotential perturbations affect temperature, humidity, and wind, although terrain and below-ground extrapolation may explain part of this pattern. The audit can guide input-quality checks for forecasts used by early-warning services. It measures model responses rather than atmospheric causality; event-level benefits remain untested.

1 Introduction

Pangu-Weather and Aurora produce global, high-resolution forecasts at low marginal cost [1, 2]. Forecasting is a core component of multi-hazard early-warning systems [3], yet aggregate RMSE and ACC do not show whether an error in one input remains local or spreads into variables used by a warning service. Data-driven weather models can score well while showing physically inconsistent structure or unrealistic growth of small perturbations [4, 5].

Controlled perturbations have produced physically plausible adjustment in Pangu-Weather under canonical dynamical tests [6], but small-amplitude perturbations have also exposed unrealistic error growth [5]. Other weather-model studies use gradients and representation analysis [7], identify advection heads in Aurora [8], inspect latent directions in GraphCast [9], or perturb atmospheric chemistry in Aurora [10]. We measure the response of every physical output and require no access to internal activations.

The paired audit uses 19×19 matrices to separate forecast displacement from verified skill change, dose sweeps to measure response size, and maps to locate added error. We test whether a replaced field returns toward the baseline rollout, which inputs produce strong cross-field responses, and whether both architectures share directional asymmetries. The measurements also select candidate field and latitude weights for a future matched retraining experiment. The climate pathway is model and input-quality assurance for forecast centres that support hazard-warning and adaptation services. We do not yet evaluate an operational decision or estimate avoided losses.

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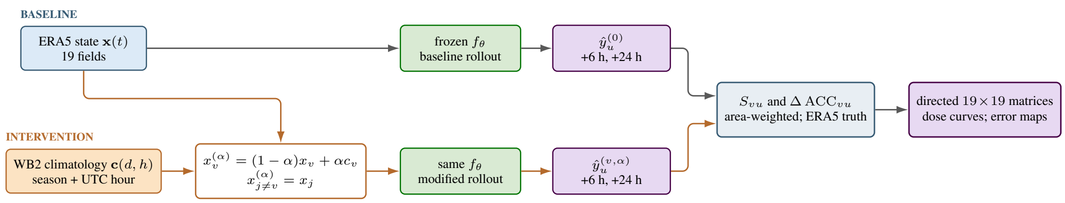


Figure 1: Paired intervention audit. The reference state and a season- and UTC-matched climatological replacement pass through identical frozen model weights. Their forecasts are compared by displacement S and verified anomaly-skill change $\Delta\,\mathrm{ACC}$. Repeating the comparison for every input, output, and dose yields directed matrices and spatial summaries.

2 Methods

ERA5 provides initial states and verification targets [11]; WeatherBench 2 climatology is matched by day of year and UTC hour and is computed over 1990–2019 [12]. We use Aurora (1.3B parameters; two input times) and Pangu-Weather (four 64M-parameter networks; one input time) on a $0.25°$ grid. The main sample has 48 initializations in 2024: four dates per month, balanced over 00/06/12/18 UTC. The fields are MSLP, 10-m u/v, 2-m temperature, and Z, Q, T, u, v at 1000, 850, and 500 hPa. The selected Aurora field is replaced at both input times. A separate seven-field Aurora check spans 2024–2025 (Appendix A.1).

Input field v is replaced by climatology. For output u, lead τ, and date set $\mathcal{D}$, we report

$$S_{vu}(\tau) = \frac{1}{|\mathcal{D}|} \sum_{t \in \mathcal{D}} \frac{\mathrm{RMSE}_w(\hat{y}_u^{(v)}, \hat{y}_u^{(0)})}{\sigma_w(y_u)}, \tag{1}$$

$$\Delta\,\mathrm{ACC}_{vu}(\tau) = \frac{1}{|\mathcal{D}|} \sum_{t \in \mathcal{D}} \Big[\mathrm{ACC}_w(\hat{y}_u^{(v)}, y_u, c_u) - \mathrm{ACC}_w(\hat{y}_u^{(0)}, y_u, c_u) \Big]. \tag{2}$$

Spatial sums use normalized $\cos\phi$ weights. S measures distance from the baseline forecast; negative $\Delta\,\mathrm{ACC}$ records a loss of anomaly skill against ERA5. For the dose sweep, $Z1000$ is mixed with climatology at $\alpha \in \{0, .2, .4, .6, .8, 1\}$; spatial composites compute temporal RMSE at each grid cell. The replacement retains season and UTC hour but creates an inconsistent multivariate state. Thus $\alpha = 1$ is a finite stress test, not a local gradient or evidence of atmospheric causality.

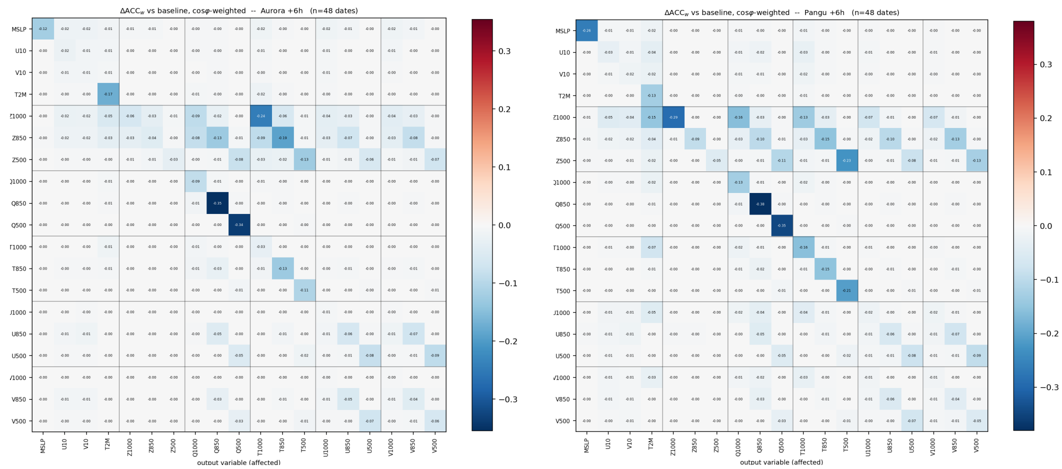
The two metrics separate model sensitivity from verified skill. Large S_{vu} with small $|\Delta\,\mathrm{ACC}_{vu}|$ means that output u moved substantially while retaining its spatial anomaly pattern against ERA5. A small S_{vu} can occur when the remaining channels reconstruct input v. We therefore interpret each diagonal entry together with the rest of its input row.

3 Results

Direction and attenuation. The matrices are asymmetric. At +6 h, $S_{Z850,V850}$ is 0.38 for Aurora and 0.47 for Pangu, while the reverse entries are 0.04 and 0.05. The roughly ninefold contrast recurs in both architectures. Diagonal $S_{Z500,Z500}$ falls from 0.08 to 0.04 in Aurora and from 0.19 to 0.09 in Pangu between +6 and +24 h. The corresponding +24/+6 ratios are 0.50 and 0.47. They measure motion toward the baseline trajectory; lower error against ERA5 has not been established, and off-diagonal responses persist.

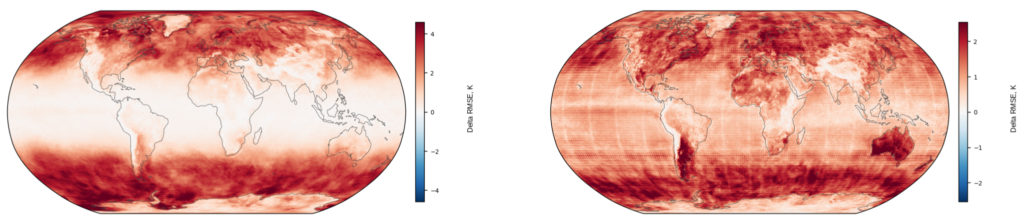
We call this partial internal reconstruction because the comparison target is the baseline forecast rather than truth. By the time $Z500$ approaches that forecast, the intervention has already changed wind and other outputs. The audit separates reconstruction of the replaced channel from stability of the full multivariate state.

Dose and spatial footprint. All six $Z1000$ doses rise without a plateau (Appendix Figs. A5–A8). The $T1000$ response slope in Aurora falls from 0.690 to 0.217 between +6 and +24 h, while Pangu

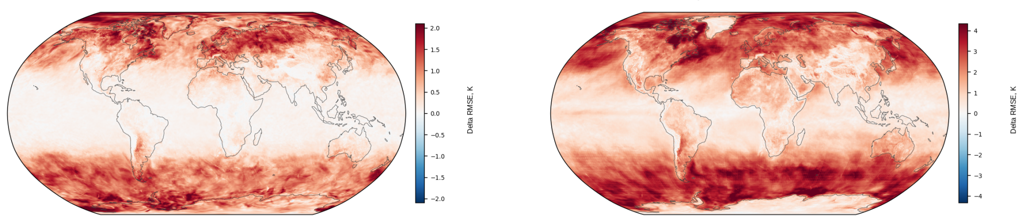


(a) Aurora, $\Delta\,\mathrm{ACC}_w$, +6 h (b) Pangu-Weather, $\Delta\,\mathrm{ACC}_w$, +6 h

Figure 2: Skill loss at +6 h: rows are replaced inputs and columns are outputs. Negative $\Delta\,\mathrm{ACC}_w$ means degradation. Panel scales differ, so compare printed values. Full matrices appear in Appendix Figs. A1–A2.



(a) Aurora, +6 h: $6.02\times$ (b) Pangu-Weather, +6 h: $3.35\times$



(c) Aurora, +24 h: $2.01\times$ (d) Pangu-Weather, +24 h: $3.06\times$

Figure 3: Change in temporal $T1000$ RMSE after replacing $Z1000$; red denotes larger error. Labels give mean per-cell ratios without area weighting.

rises from 0.406 to 0.614. Damping depends on the model and lead. After full replacement, mean per-cell $T1000$ RMSE increases by $6.02/2.01\times$ in Aurora and $3.35/3.06\times$ in Pangu at +6/+24 h. The increase covers 99–100% of cells. The maps show different geographic patterns, including latitude-dependent bands.

4 Discussion

Measured behavior. Diagonal attenuation and cross-field response occur in the same rollout. Both models reconstruct part of $Z500$ from context after the inconsistent input has changed other outputs. Replacing $Z1000$ and $Z850$ produces substantial responses in temperature, humidity, and wind. Terrain imprint and below-ground extrapolation may contribute, although the experiment cannot separate them from vertical coupling. Off-diagonal matrix sums can rank inputs by outward influence, $\Delta\,\mathrm{ACC}$ separates displacement from skill loss, and the maps locate accumulated error. A matched

Table 1: Point estimates over 48 dates. Confidence intervals have not yet been computed.

Diagnostic	Aurora	Pangu-Weather
Response of $V850$ to $Z850$; response of $Z850$ to $V850$, +6 h	$0.38;\ 0.04$	$0.47;\ 0.05$
$Z500$ diagonal, +6/+24 h	$0.08/0.04$	$0.19/0.09$
$T1000$ response to $Z1000$, +6/+24 h	$0.690/0.217$	$0.406/0.614$
Per-cell $T1000$ RMSE, +6/+24 h	$6.02\times/2.01\times$	$3.35\times/3.06\times$

study could use these quantities to propose λ_N and $w_N(\phi)$ in $\mathcal{L} = \sum_N \lambda_N \sum_s w_N(\phi_s)\ell_N(s)$; the present study does not test those weights.

Limitations. The main matrices contain point estimates from 48 dates in 2024 and two leads. Month-block intervals have not yet been computed; Appendix A.2 specifies the paired resampling test. The seven-field Aurora audit over 2024–2025 retains both the low $Z500$ diagonal and the stronger response of $V850$ to a $Z850$ intervention (Appendix Fig. A3), but its field set differs and it provides no second-year Pangu result. Full climatology replacement creates a multivariate state outside the training distribution. Each cell is an interventional response from a separate rollout; the array is neither a correlation matrix nor an atmospheric causal graph. We also have not stratified results by hazard type or connected field errors to warning thresholds, so the climate pathway remains prospective.

The current S diagonal establishes convergence toward the unperturbed forecast. Correction against truth requires $D_u^{(v)}(\tau) = \mathrm{RMSE}_w(\hat{y}_u^{(v)}, y_u) - \mathrm{RMSE}_w(\hat{y}_u^{(0)}, y_u)$ at three or more leads. Recovery would require decreasing $D_v^{(v)}$; positive $D_u^{(v)}$ for $u \neq v$ would measure the cost paid by neighboring fields. The lower-level result also needs a below-ground mask, terrain-height strata, and a season- and UTC-matched anomaly replacement to separate model behavior from the construction of the intervention.

Future work and climate relevance. Paired month-block intervals are needed for the directional contrast and attenuation ratios. The ERA5-referenced correction test will cover +6, +12, +24, and +48 h. A matched retraining study will derive candidate field weights from off-diagonal influence and spatial weights from the mapped latitude profiles, while keeping data, compute, and initialization seeds fixed. Tests on GenCast, FuXi-S2S, and LaDCast will pair ensemble members through common random seeds and report S, $\Delta\,\mathrm{ACC}$, CRPS, and spread–skill behavior. Mid-rollout ERA5 replacement remains a retrospective test of correctable response paths.

The near-term user is a forecast-centre team validating a data-driven model for a regional hazard-warning service. An input with a reproducibly large, skill-degrading response would be flagged for stricter quality control, targeted checks of redundant observations, or separate calibration. Success means lower truth-relative error and fewer threshold errors on the service's event set without degrading other fields or regions. Because field and latitude weighting can redistribute errors, a retraining study must report every output and the relevant regional slices.

Conclusion. Paired replacement exposes cross-field responses hidden by aggregate scores. In both models, $V850$ responds much more strongly to replacement of $Z850$ than $Z850$ responds to replacement of $V850$. Both models partly reconstruct $Z500$, while other fields remain displaced, and lower-level geopotential perturbations affect several outputs. The next required tests are month-block uncertainty, truth-relative recovery at four leads, and region- and event-specific validation with a warning service.

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A Supplementary diagnostics and evaluation provenance

A.1 Data splits and matrix conventions

The repository contains several evaluation sets. Its ERA5 cache and broad forecast-verification campaign cover 2024–2025. The paired results in the main paper use a narrower common set: 48 dates from 2024 for both models, all 19 fields, and leads of +6 and +24 h. A separate Aurora-only audit uses seven fields and 48 dates split across 2024–2025. We report that audit below as a temporal check and do not pool it with the two-model estimates.

Every matrix uses rows for replaced inputs and columns for affected outputs. $S_{vu} \geq 0$ measures displacement from the unperturbed forecast after normalization by the verified field scale. A large diagonal entry means that the model cannot reconstruct that input from the other 18 fields; an off-diagonal entry records propagation into another output. $\Delta\,\mathrm{ACC}_{vu} < 0$ means that the same intervention reduces anomaly skill against ERA5. Color limits differ by model, lead, and metric. Cross-panel comparisons therefore use the cell values, not hue.

A.2 Statistical checks before submission

The resampling unit is the initialization date, not a matrix cell. Cells share input states and are not independent observations. The added `analysis/bootstrap_19var.py` script groups the four dates in each calendar month, resamples the 12 month blocks with replacement, and applies the same sample to Aurora and Pangu. Each of 5000 replicates recomputes the mean matrices. The outputs are 95% intervals for every cell, the difference $S_{Z850,V850} - S_{V850,Z850}$, the diagonal ratio $S_{Z500,Z500}(24)/S_{Z500,Z500}(6)$, and the area-weighted cross-model correlation.

The current figures contain point estimates. An earlier comparison script reported $r = 0.913$ from unweighted `rel_sens`, whereas the paper displays `rel_sens_w`. The script now defaults to area-weighted metrics, and the cross-model correlation is withheld until the source tables are rerun. The generated `weighted_matrix_ci.csv` and `key_results_bootstrap.csv` should be the sole numerical source for the final manuscript.

Attenuation in S measures return toward the baseline trajectory. A claim of correction against truth additionally requires

$$D_u^{(v)}(\tau) = \mathrm{RMSE}_w(\hat{y}_u^{(v)}(\tau), y_u(\tau)) - \mathrm{RMSE}_w(\hat{y}_u^{(0)}(\tau), y_u(\tau)). \tag{3}$$

A decline in $D_v^{(v)}$ across leads together with positive $D_u^{(v)}$ for $u \neq v$ would demonstrate recovery of the replaced field and transfer to its neighbors. This test should use at least three leads; the stored +6 and +24 h results provide only two points.

Two controls separate model structure from the construction of the replacement. For lower pressure levels, grid cells where surface pressure is below 1000 or 850 hPa should be excluded, with lowlands and high terrain reported separately. To test distribution shift, climatology should be replaced by an anomaly from another date matched by season and UTC, optionally scaled to the same perturbation norm. The conclusions about asymmetry and the leading inputs remain credible only if their signs and matrix ranks persist.

A.3 Nineteen-field response matrices

In both models, replacement of $Z850$ and $Z500$ changes several wind outputs, whereas replacing wind usually produces a smaller response in geopotential. The lower panels show whether the displaced forecasts also lose verified skill. Most entries are negative, but their magnitude is not proportional to S because a displaced forecast can retain spatial anomaly correlation.

A.4 Aurora seven-field audit, 2024–2025

This check tests whether the ordering of selected effects persists over time. It uses a different field set and unweighted spatial statistics, so its values cannot be pooled with the main sample or treated as an independent Pangu replication. Across dates from two years, the $Z500$ diagonal remains low and the response of $V850$ to replacement of $Z850$ remains stronger than the reverse response.

A.5 Candidate field and latitude weights

Each input yields a retained diagonal response R_M and total off-diagonal influence I_M. A latitude profile $E_{MN}(\phi)$ can be computed from the grid-cell composites. R_M identifies fields partly reconstructed by the model, while I_M measures how widely an input perturbation spreads. $E_{MN}(\phi)$ would show where a common $\cos\phi$ weight misses output-specific error. Candidate weights must be normalized and fixed before a matched training comparison with identical data, optimization budget, and initialization seeds.

A.6 Ensemble audit and mid-rollout field correction

For probabilistic systems, baseline and modified ensemble members should share random seeds. GenCast samples diffusion-based 15-day ensembles [13]; FuXi-S2S generates 42-day ensembles with learned, flow-dependent perturbations [14]; LaDCast generates hourly ensembles through latent diffusion [15]. Evaluation must report the distributions of S and $\Delta\,\mathrm{ACC}$, CRPS, spread–skill behavior, and the ensemble mean.

A second experiment should replace one predicted field with verifying ERA5 at +6, +12, +24, or +48 h and continue the rollout. Reduced downstream error would locate a correctable transfer pathway; renewed error in the replaced field would measure recurrence. This retrospective reference-replacement experiment is unavailable in an operational forecast. Repeating it for each field and lead would separate errors that the model damps, errors that recur, and errors that contaminate neighboring fields.

A.7 $Z1000$ dose response

The dose panels show mean normalized error at six measured values of α and a least-squares slope through those points. The intercept includes baseline forecast error, and the slope describes the finite path to climatology. A slope measured for $Z1000$ does not quantify the effect of an arbitrary noise fraction in another input field. The opposing lead-time changes in Aurora and Pangu show that the coefficient is model-specific.

A.8 Verified skill rows and spatial composites

The $\Delta\,\mathrm{ACC}$ rows test whether forecast displacement also reduces anomaly skill. Spatial composites show where added temporal RMSE occurs; they are not signed bias maps. The fractions of degraded cells and RMSE ratios below are unweighted grid-cell summaries and must be distinguished from the area-weighted global metrics used in the matrices.

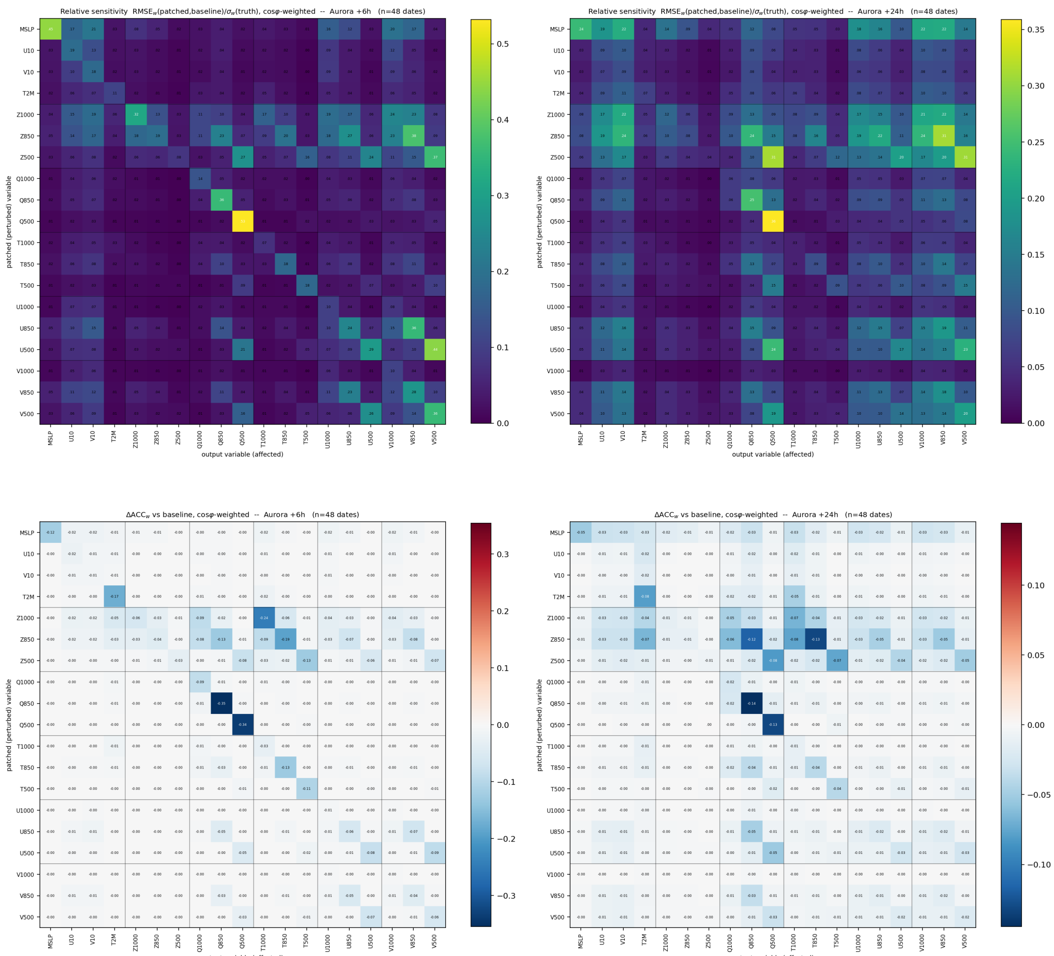

Figure A1: Aurora response matrices over 48 dates in 2024. Top: area-weighted sensitivity S; bottom: Δ ACC; left: +6 h; right: +24 h. At +6 h, the response of $V850$ to replacement of $Z850$ is 0.38; the reverse response is 0.04. The MSLP and $Z500$ diagonal entries (0.45 and 0.08) separate unique input influence from contextual reconstruction. Negative values in the lower panels are losses of verified anomaly skill.

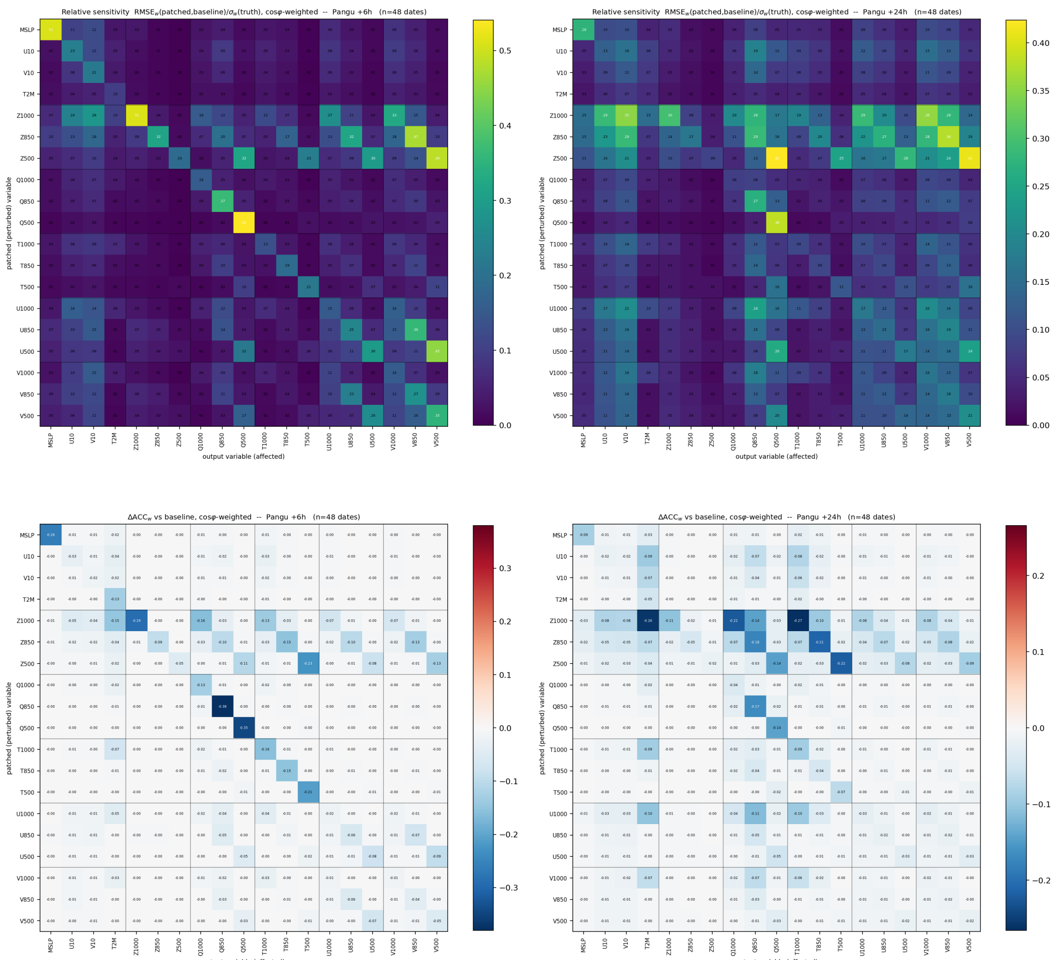

Figure A2: Pangu-Weather response matrices over the same 48 dates. Panel order and metrics match Fig. A1. At +6 h, the response of $V850$ to replacement of $Z850$ is 0.47; the reverse response is 0.05. MSLP and $Z500$ diagonal sensitivities are 0.53 and 0.19. The same dominant responses occur in both models, but their magnitudes differ.

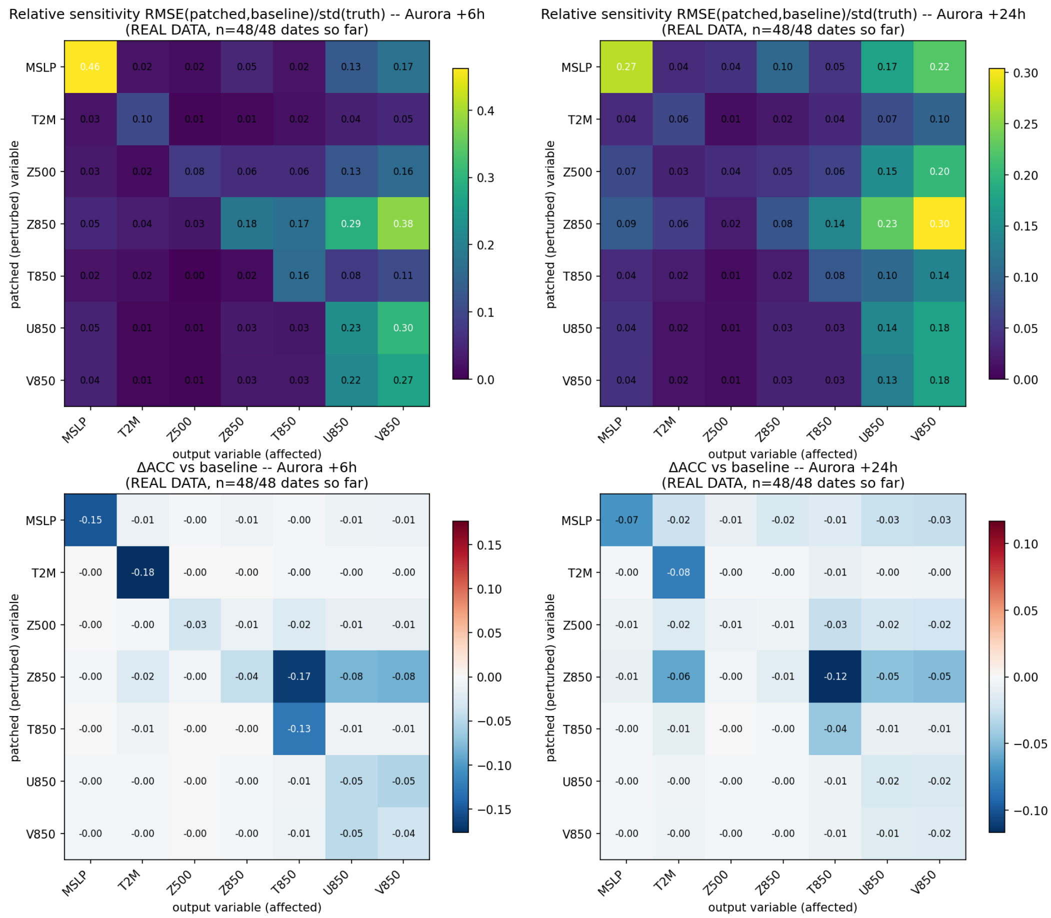

Figure A3: Two-year Aurora check on seven fields and 48 dates spanning 2024–2025. These panels use unweighted spatial statistics and are not pooled with the area-weighted 19-field results. The +6 h diagonal ordering persists: MSLP sensitivity is 0.46 and $Z500$ sensitivity is 0.08; at +24 h they are 0.27 and 0.04. The response of $V850$ to replacement of $Z850$ remains stronger than the reverse response. This check does not extend the Pangu comparison beyond 2024.

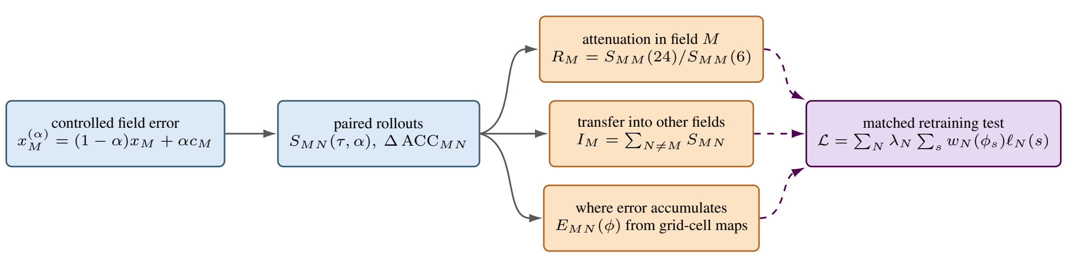

Figure A4: From measured responses to a loss-weighting experiment. Solid arrows show outputs of the audit: a controlled dose yields attenuation, cross-field responses, and grid-cell error maps. Dashed arrows mark an unvalidated design step. Influence ranks may set field weights λ_N, while latitude profiles computed from the maps may set field-specific spatial weights $w_N(\phi)$. Improvement requires matched retraining against the original loss.

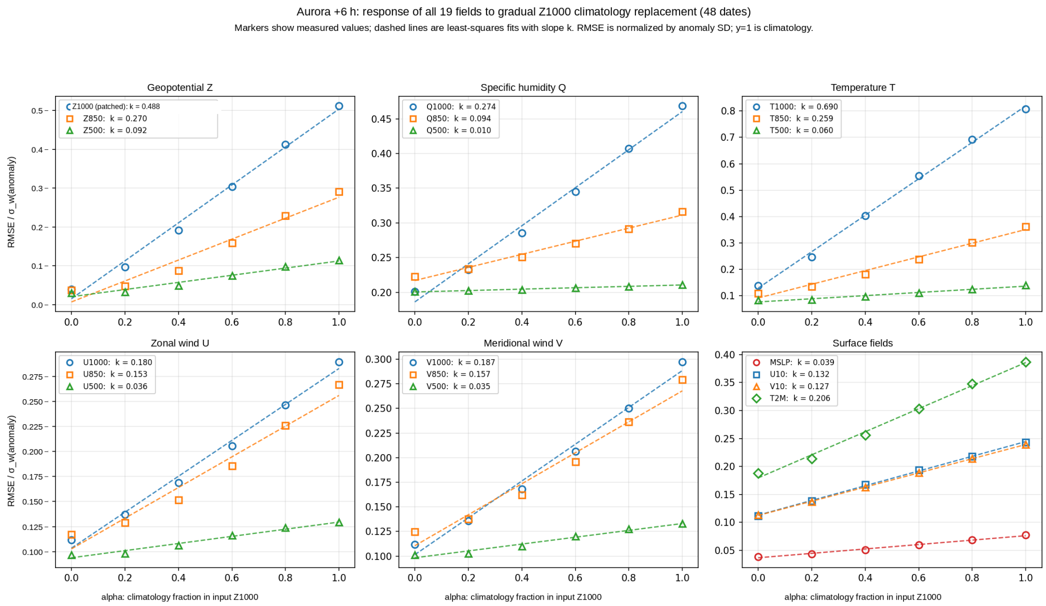

Figure A5: Aurora at +6 h, 48 dates in 2024. Markers show the six measured climatology fractions and dashed lines show least-squares fits. The response has no plateau before $\alpha = 1$. $T1000$ has the steepest slope (0.690), followed by $Z1000$ (0.488) and $Q1000$ (0.274). The y-axis is RMSE normalized by anomaly standard deviation; one is the climatological no-skill scale.

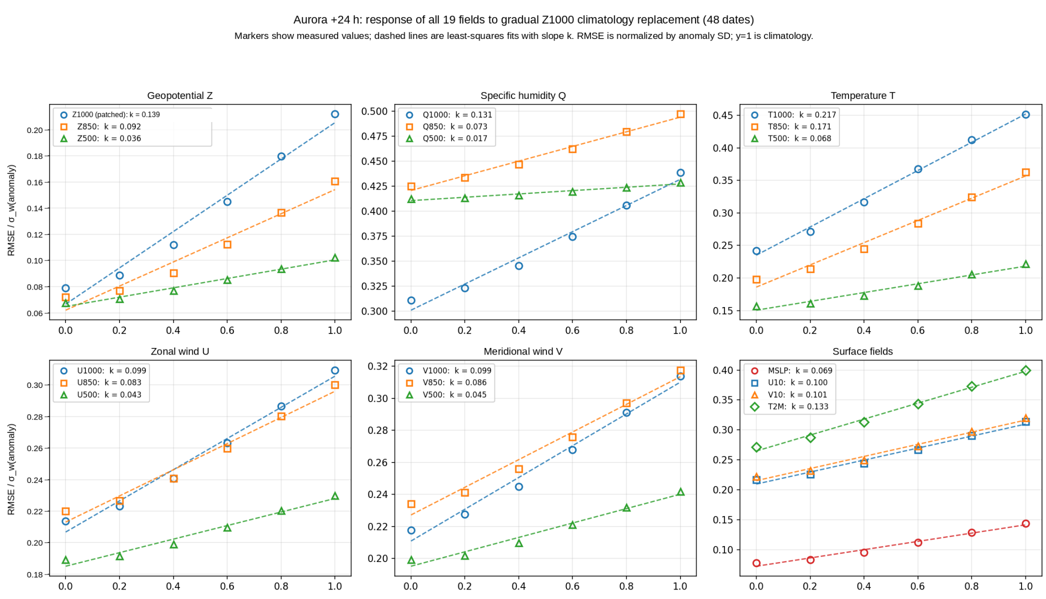

Figure A6: Aurora at +24 h on the same dates and dose grid as Fig. A5. The $T1000$ slope falls to 0.217 and the $Z1000$ slope to 0.139, while the $Q1000$ slope is 0.131. The nonzero intercepts are the baseline forecast errors, not intervention effects.

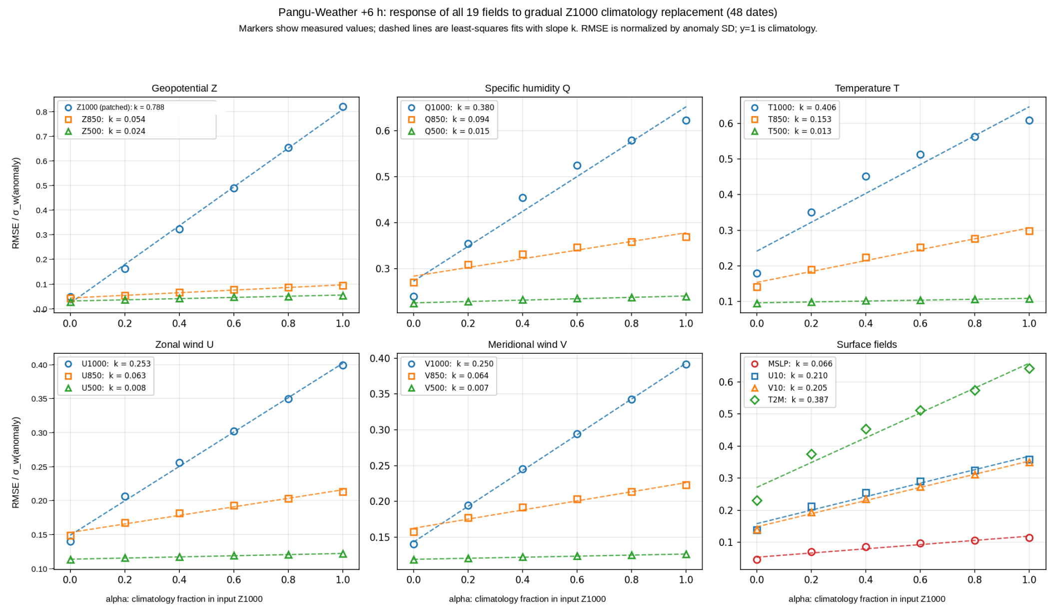

Figure A7: Pangu-Weather at +6 h, 48 dates in 2024. The $T1000$ slope is 0.406, below Aurora's 0.690, whereas the lower-level wind and $T2M$ responses are steeper. This is a measured finite path from the observed input to climatology; the fitted line summarizes these six points and is not a local derivative.

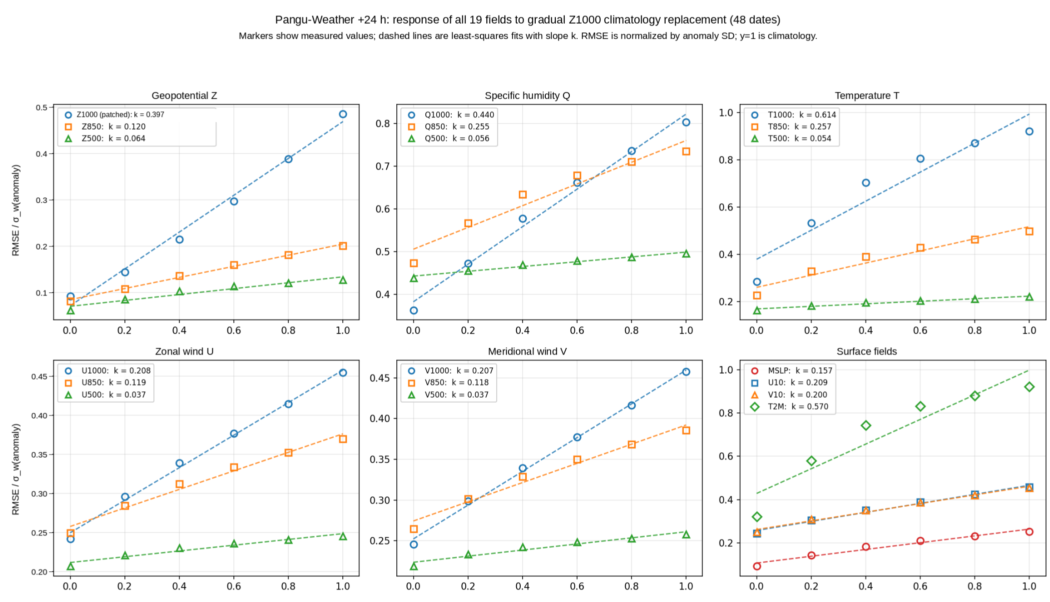

Figure A8: Pangu-Weather at +24 h on the same dates and dose grid as Fig. A7. The $T1000$ slope rises to 0.614 and the $T2M$ slope to 0.570. Aurora weakens over the same lead change, so the difference is in lead-time behavior rather than only scale.

Figure A9: Δ ACC rows after replacing $Z1000$, $Z850$, and $Z500$ in both models, using the common 48-date 2024 set. The strip isolates the mass-field interventions from the full matrices. Negative cells record skill loss; near-zero cells mean that the anomaly pattern remains correlated with ERA5 even when the forecast itself moves. Each row retains its parent-matrix scale, so compare printed values rather than color saturation.

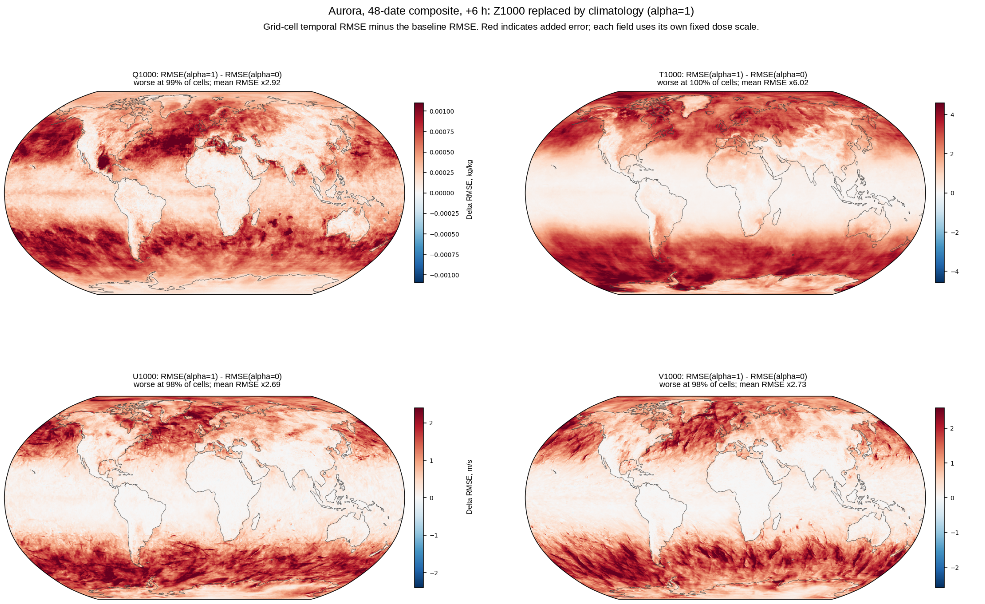

Figure A10: Aurora spatial composite after full $Z1000$ replacement ($\alpha = 1$, +6 h; 48 dates in 2024). Each map shows temporal RMSE for the modified forecast minus temporal RMSE for the baseline at a grid cell. Red therefore marks added error, not a signed forecast bias. $T1000$ has a $6.02\times$ global mean per-cell RMSE ratio, and the increase covers nearly the full grid.

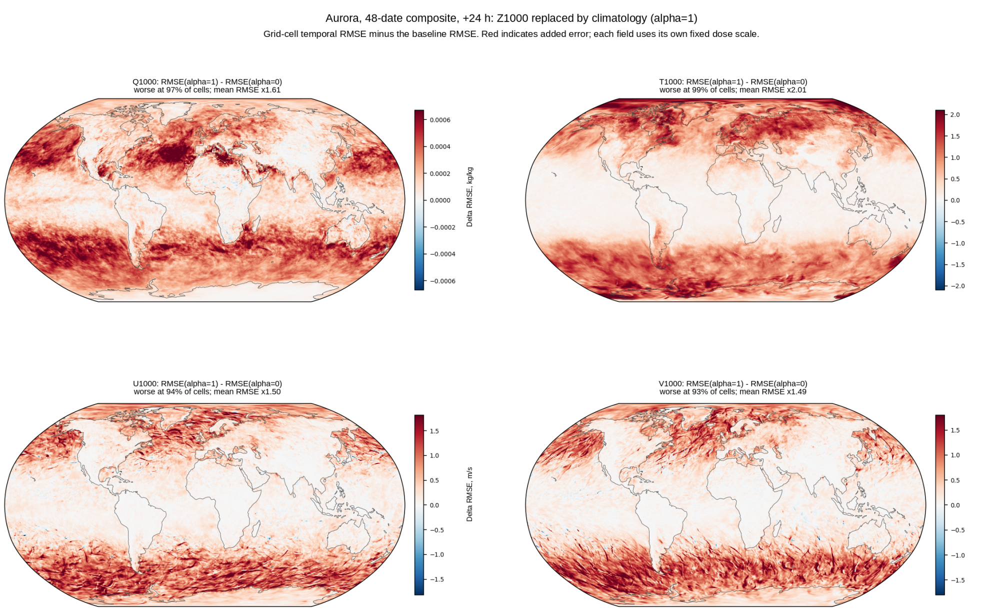

Figure A11: Aurora spatial composite at +24 h. The response is weaker than at +6 h: the $T1000$ global mean per-cell RMSE ratio falls from $6.02\times$ to $2.01\times$. The four displayed fields still worsen over 93–99% of cells, so the smaller global mean is not caused by a small isolated region.

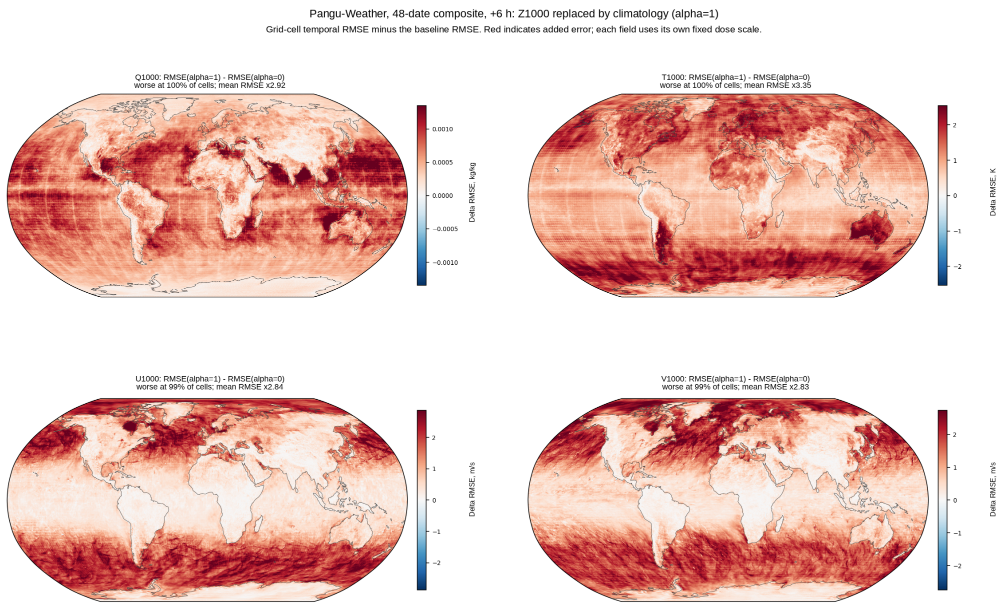

Figure A12: Pangu-Weather spatial composite at +6 h. The four displayed fields worsen over 99–100% of cells. For $T1000$, the global mean per-cell RMSE ratio is $3.35\times$, lower than Aurora at this lead, but the map remains broadly positive across latitude bands.

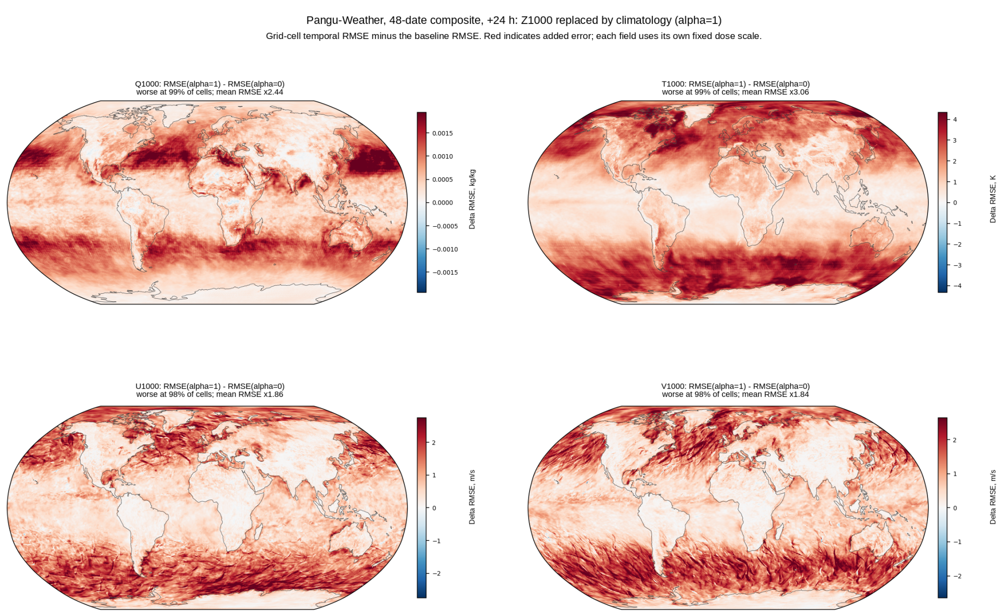

Figure A13: Pangu-Weather spatial composite at +24 h. $T1000$ retains a $3.06\times$ global mean per-cell RMSE ratio, close to its +6 h value and above Aurora's $2.01\times$ at this lead. This persistent response is consistent with the Pangu dose slopes, which grow rather than shrink from +6 to +24 h.